NETWORK DEVICES

Course: Diploma in Cyber Security

Topic: Network Devices

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Introduction

Network devices help computers, phones and other devices communicate with each other and access the internet. Common devices include routers, switches, hubs and access points. In this assignment, I have shown a simple home network and explained the role of these devices.

Task 1: Simple Home Network Diagram

A simple home network can use a router as the main network device. A switch can provide extra wired ports, while phones and smart TVs can connect through Wi-Fi.

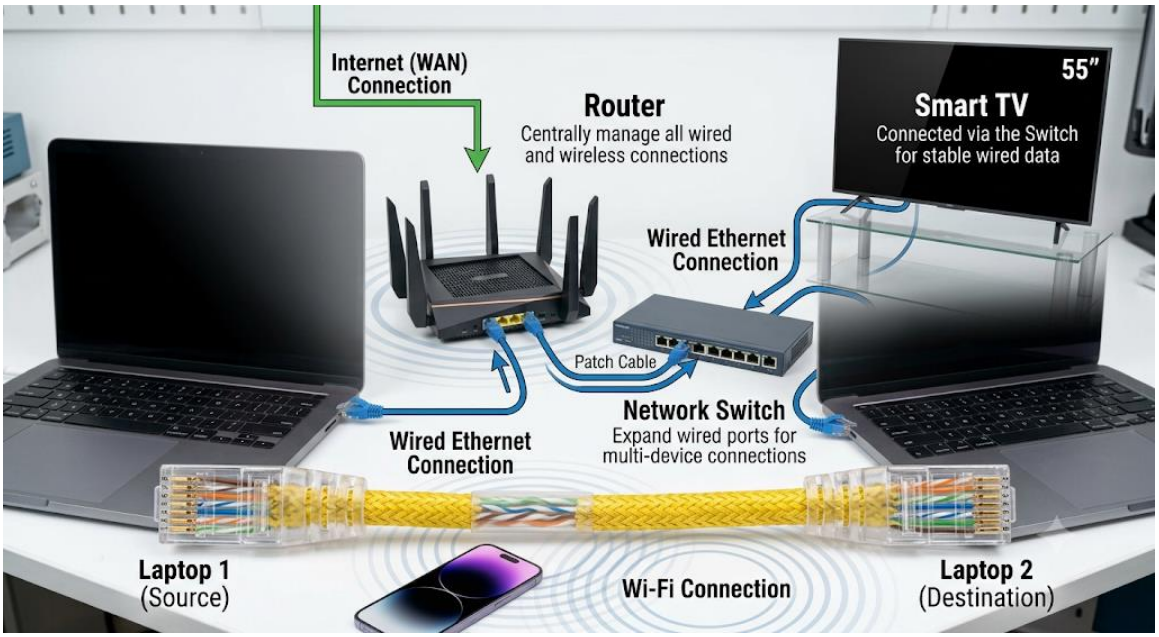


Figure 1: Simple Home Network with Router, Switch and Three Devices

Task 2: Comparison of Network Devices

Network Device	Real-World Use	Daily App / Service Example
Router	Connects a local network to the internet and directs traffic.	Wi-Fi for YouTube streaming on a phone or smart TV.
Switch	Connects multiple wired devices inside a LAN.	LAN gaming where multiple PCs or consoles use wired connections.

Hub	**Sends incoming data to all connected ports.**	**Mainly useful as a simple learning example or in older basic networks.**
Access Point	**Provides Wi-Fi access and connects wireless users to a wired network.**	**Phone connecting to Wi-Fi for Instagram, WhatsApp or web browsing.**

Task 3: Ordering Food on Zomato

When I order food on Zomato, my phone first connects to an access point if I am using Wi-Fi. The access point sends the data to the network, and the router forwards it towards the internet. In a larger network, switches can connect different wired devices. The request then travels through the internet to the Zomato server, and the response comes back through the network to my phone.

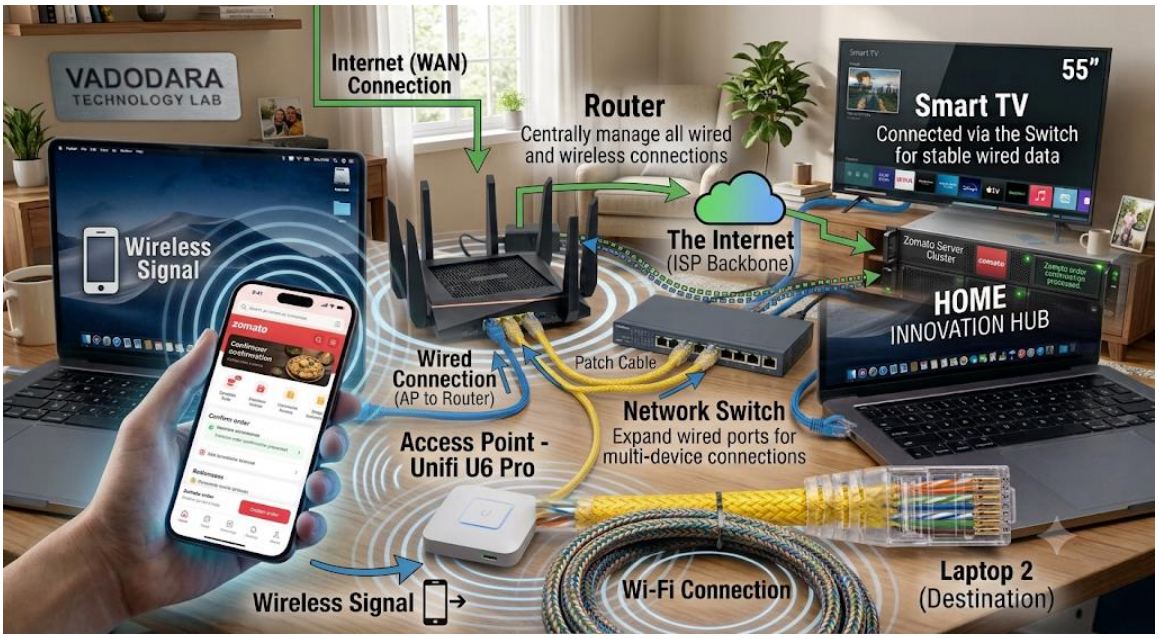


FIGURE 2

Task 4: Gaming PC Far from Wi-Fi Router

One option is a powerline adapter. One unit connects to the router by Ethernet and another unit is placed near the gaming PC, then the PC connects by Ethernet. Another option is a Wi-Fi extender or mesh unit with an Ethernet port. It receives Wi-Fi from the router and gives a wired Ethernet connection to the gaming PC.

Option 1: Powerline Adapter

Router -> Powerline Adapter -> Home electrical wiring -> Second Powerline Adapter -> Ethernet cable -> Gaming PC

Option 2: Wi-Fi Extender / Mesh Unit

Router -> Wi-Fi -> Extender/Mesh Unit -> Ethernet cable -> Gaming PC

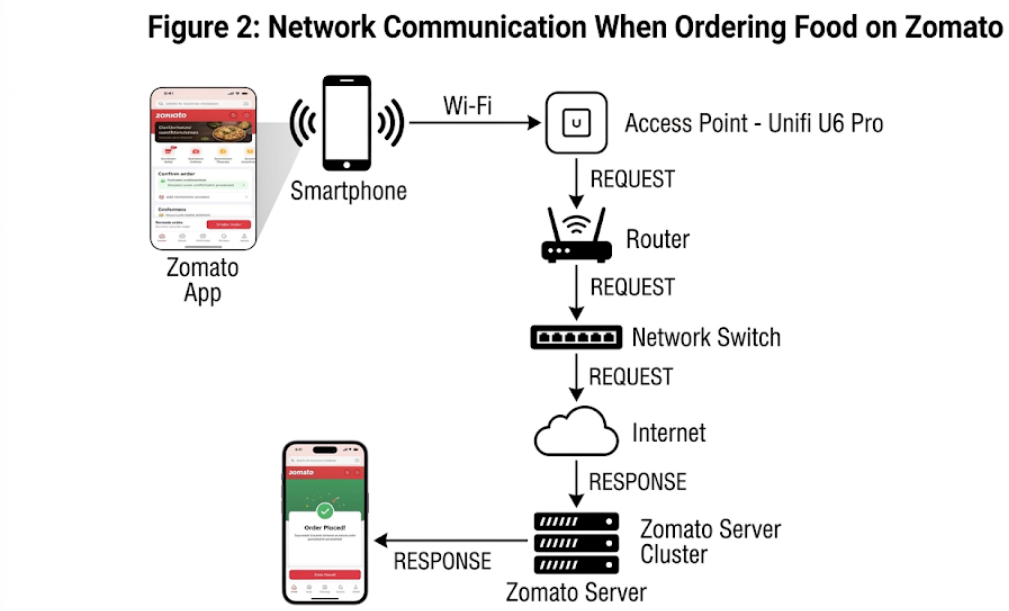


Figure2:

Conclusion

In this assignment, I learned how common network devices work together. A router connects networks, a switch connects wired devices, a hub repeats data to all ports, and an access point provides Wi-Fi. Choosing the right device depends on the distance, number of devices and type of connection needed.

Quick Revision

- Router = connects networks and directs traffic.
- Switch = connects wired devices in a LAN.
- Hub = sends data to all ports.
- Access Point = provides Wi-Fi access.
- Powerline adapter = can carry network data over suitable home electrical wiring.
- Wi-Fi extender / mesh with Ethernet = can provide a wired port farther from the main router.